

## Clinical profile and outcome of early surgery in neonatal-onset glaucoma presenting over a 5-year period.

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Neonatal-onset glaucoma (NOG) is a severe form of childhood glaucoma and is not always due to primary congenital glaucoma (PCG). Due to advances in neonatal care, the incidence of NOG is rising, but it remains an under-reported entity. The objective of the paper was to study the clinical profiles, surgical and visual outcomes of NOG at least 1 year following early surgery. Prospective interventional cohort study at a tertiary care referral centre. Babies with NOG, who presented between January 2013 and December 2017, had a history suggestive of disease onset within 1 month of birth, and underwent surgery by 3 months of age, were prospectively enrolled. Those who completed a 1-year follow-up after surgery were analysed. 94 eyes of 53 babies were analysed. 35 (66%) had PCG. Neonatal congenital ectropion uveae, congenital rubella syndrome, Peter's anomaly and Sturge-Weber syndrome comprised the non-PCG group. The mean age at presentation and surgery was  $24.8 \pm 21.9$ , and  $36.7 \pm 29.9$  days. Additional glaucoma surgery was required in 43 of the 94 eyes (45.7%). PCG had significantly better outcomes than other glaucomas at all time points. 28.3% of eyes had good vision (LogMar (0-0.5)), 34.7% had moderate visual impairment (LogMar 0.7-1.0) and 16% were blind (LogMar  $<1.62$ ). Our study shows that NOG does not always have a dismal prognosis. A small but significant proportion could have other underlying conditions than PCG. Timely surgery and rigorous amblyopia therapy resulted in good outcomes in terms of intraocular pressure control and vision in this cohort.

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## Primary congenital glaucoma.

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Primary congenital glaucoma is a rare but potentially devastating disorder that family physicians must recognize as early as possible in infants and young children. Symptoms, which may not be present at birth, classically include photophobia, excessive tearing and blepharospasm. The cornea may be enlarged and hazy. Increased intraocular pressure rapidly damages the optic nerve in infants. Treatment includes surgery with regular follow-up evaluation.

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## A Neonate With Diabetes Mellitus, Congenital Hypothyroidism, and Congenital Glaucoma.

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Neonatal diabetes mellitus (NDM) is a rare condition with more than 20 monogenic genes associated with it. GLIS3 gene-encoded GLI similar protein 3, as a transcription factor, is involved in the development of the pancreas, liver, kidneys, eye, and thyroid. We report a preterm female neonate with coarse facial features and hyperglycemia, later diagnosed with neonatal diabetes mellitus, congenital hypothyroidism (CH), congenital glaucoma (CG), and renal cysts, secondary to GLIS3 gene mutation. It is a rare genetic disorder involving multiple organ systems with progressive development of symptoms requiring long-term surveillance and management.

## [Molecular basis of ameloblastoma pathogenesis: A review].

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Ameloblastoma is the most common and highly invasive benign odontogenic tumor. Its pathogenesis is not yet fully understood. To describe the most important molecular findings that promote the proliferative activity of ameloblastoma and the factors involved that encourage invasion into surrounding bone tissues. A search for scientific evidence was conducted through the following databases: Science Direct, Medline, Wiley, Web of Science, and Google Scholar. A total of 32 articles were reviewed, with inclusion criteria being articles published in English and Spanish; descriptive and analytical studies, narrative and systematic reviews published from January 2015 to June 2021. Letters to the editor were excluded. The biological molecular findings that allow ameloblastoma to invade surrounding tissues involve alterations in the RANK/RANKL/OPG pathways, transforming growth factor beta (TGF- $\beta$ ), Wnt/ $\beta$ -catenin pathway, and matrix metalloproteinases, as well as alterations in MAPK and SHH pathways that facilitate the proliferation and tumor development of ameloblastoma. These findings are fundamental for a better understanding of the pathways involved in the pathogenesis of ameloblastoma.

## Clinicopathologic Profile and Surgical Modalities in Mandibular Ameloblastoma: A Descriptive Study.

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Ameloblastoma is a benign neoplasm composed of epithelial tissue with invasive and infiltrative behavior at the local level and a high recurrence rate, with various histopathologic patterns and clinical forms. Approximately 85% of conventional ameloblastomas occur in the mandible, most often in the body, angle, and ascending ramus area. The treatment modalities include both conservative and radical treatments. Postoperative follow-up is most important in the treatment of ameloblastoma. To describe the clinicopathologic profile of mandibular ameloblastoma in patients undergoing different surgical modalities. The primary objective was to describe the clinicopathologic profile and surgical management of mandibular ameloblastoma in patients aged  $\geq 18$  years, who had reported to a tertiary dental care center for follow-up during the study period. The secondary objective was to describe the distribution of comorbidities associated with different surgical modalities and reconstructive methods. A total of 34 patients with mandibular ameloblastoma who underwent various surgical modalities between 2011 and 2021 were studied. Information was collected using a predesigned proforma and statistically analyzed. Thirty-four review cases of ameloblastoma were included in the study. The patients were analyzed concerning age, sex, site, size, clinical presentation, radiographic pattern, histopathologic subtype, type of surgery, and associated comorbidities. Most cases of mandibular ameloblastoma involve the age of 16 to 55 years. The mean age of occurrence was found to be  $35.5 \pm 13.2$ . A female preponderance, a tumor size range of 2 to 4 cm, a multicystic variant, involvement of the mandibular body in the premolar-molar area, root resorption, cortical perforation, and a follicular type of histopathologic pattern were the common presentations. Isolated anterior tumors restricted to the incisor/canine region were not found. The common surgical modalities undertaken were conservative methods such as enucleation, and chemical cauterization, and radical methods such as marginal mandibulectomy and segmental resection. Reconstruction using a titanium plate or free fibular graft was performed in the indicated cases. The common comorbidities included difficulty in chewing and loss of facial contour. Recurrence after surgical treatment was rare. Only 9% of cases developed a recurrence within 5 years. No recurrence was noted in cases treated with radical treatment, whereas 50% of cases treated with conservative methods showed recurrence. The age of occurrence, site, and size of the tumor, cortical perforation, root resorption,

histopathologic type, and radiographic patterns are widely considered factors in devising a treatment plan for mandibular ameloblastoma. However, there may be rare instances where these tumors behave differently regardless of their innocuous clinicopathologic presentation. Surgical procedures such as segmental resection and marginal mandibulectomy were found to be promising for the eradication of the tumor, and prevention of recurrences and metastasis. However, conservative measures such as enucleation and chemical cauterization were fraught with an increased risk of tumor recurrence and metastasis. Future studies with a larger sample size should focus on the clinicopathologic characteristics of ameloblastoma to elucidate its varied behavior and develop newer and advanced treatment modalities that would provide better surgical and postsurgical outcomes in affected patients.

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## "Is it time for personalised medicine for Ameloblastoma?": A hypothesis.

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Ameloblastoma is a benign odontogenic tumor that is locally destructive. The most common treatment option is surgery, which often results in disfigurement of the face. BRAFV600E is the common gene mutation associated with its pathogenesis. Therefore, this paper hypothesizes the use of targeted drug therapy against this mutated gene.

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## Ameloblastic carcinoma, secondary type: a case report.

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Malignant variants of ameloblastoma include metastasizing ameloblastoma, which microscopically appears benign but has metastasized and ameloblastic carcinoma that exhibits malignant histopathologic features. Ameloblastic carcinoma is classified into 2 types: a primary odontogenic malignancy and a secondary type resulting from malignant transformation of ameloblastoma. Most secondary ameloblastic carcinomas result from malignant transformation of a primary lesion after repeated postsurgical recurrences. Therefore it is rare to find an untreated secondary type presenting with histologic features of malignant transformation from an earlier benign lesion. We experienced a rare case of ameloblastic carcinoma, secondary type which might arise in an untreated ameloblastoma. The mechanism by which a preexisting benign ameloblastoma goes through a malignant transformation is also described.

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## Huge ameloblastoma of jaw-A case report.

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Ameloblastoma is a tumor of odontogenic epithelium. It is a tumour of intermediate malignant potential which lies in the gray zone between benign and malignant neoplasm. A huge ameloblastoma revealing benign cytological features in FNAC is being reported. Ameloblastoma arises from odontogenic epithelium. This tumor can occur at any age. Though traditionally divided as solid and cystic, nearly all ameloblastomas show some cystic change. This tumor shows invasive property and a remarkable tendency of recurrence. The cases showing distant metastasis are recognized as malignant ameloblastoma. Ameloblastic carcinoma is a tumor with microscopic features of ameloblastoma that

displays malignant features at cytological level.([2]) It usually has aggressive course. A case of large ameloblastoma with slow clinical course and benign cytological as well as histological features is being reported.

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## Genomic and Transcriptomic Analysis of Ameloblastoma Reveals Distinct Molecularly Aggressive Phenotypes.

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Ameloblastoma (AM) is a benign but locally infiltrative epithelial odontogenic neoplasm of the jawbones that may reach grotesque proportions and be highly recurrent if inadequately removed. The BRAFV600E mutation has been demonstrated as a key molecular event in its development; nevertheless, there are many queries about its etiopathogenesis, which are yet to be answered. In this study, we aimed to integrate the results from whole-exome sequencing (WES) and RNA sequencing in AM samples to identify novel candidate genes that may be relevant to its pathogenesis. Thirteen-matched tumors were subjected to WES and RNA-seq, respectively, to detect gene mutations and gene expression profiles, along with the presence of gene fusions. Mutations were validated using Sanger sequencing, whereas transcriptome results were validated using qPCR. The results from both molecular techniques were merged in order to identify novel candidate genes that were biologically validated with immunohistochemistry. BRAFV600E mutation was present in 62% of the analyzed cases, and each AM presented at least 2 or 3 mutations affecting cancer-driver genes. RNA-seq showed different molecular subgroups associated with an aggressive and cancer-related phenotype (epithelial-mesenchymal transition and KRAS gene sets). No gene fusions were detected among the cases. CDH11 and TGM2, novel genes associated with epithelial-mesenchymal transition in AM, were selected and validated in tissues. Both WES and RNA-seq results showed gene alterations related to proliferation, cell differentiation, and metabolic processes. These results show that AM shares many of the hallmarks of cancer secondary to the presence of oncogenic mutations or activation of oncogenic signaling pathways.

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## Difference in survival and prognosis between malignant tumors of odontogenic origin.

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The prognosis of metastatic ameloblastoma (METAM), now defined as a benign tumor, and malignant odontogenic tumors (MOT) is poorly studied in the literature. The aim of this study was to determine the prognosis and factors influencing the survival of these patients. Using the SEER database, we retrieved clinical data of patients with malignant tumors of dental origin between 1975 and 2020. They include 3 histological groups: metastatic ameloblastoma (METAM), malignant odontogenic tumors (MOT, including odontogenic carcinoma, odontogenic sarcoma, primary intraosseous carcinoma, and ameloblastic carcinoma) and ameloblastic fibrosarcoma (AFS). On 251 patients, we observed no significant difference in disease-specific survival (DSS) between the three histological groups, with 5-year DSS of 77.4 % (59.9 - 100), 84.0 % (78.7 - 89.7) and 71.1 % (50.7 - 99.7) for METAM, MOT and AFS respectively ( $p = 0.460$ ). In the MOT group, using a multivariate analysis, surgical treatment appeared to be a protective factor for DSS (HR = 0.483 [0.243 - 0.960],  $p = 0.038$ ) after taking into account the potential confounding factors mentioned above. No variable significantly influenced DSS in the METAM and AFS group in univariate analysis. We found no difference in specific survival between the three histological groups. Although

metastatic ameloblastoma was reclassified as a benign tumor by the WHO in 2017, the vital risk for patients remains consequent, and statistically similar to malignant odontogenic tumors.

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